THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 8: **Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the sequence $t_n = 2 - 5n$, the value of t_6 is: 1

(a) −27 **(b)** −28 **(c)** −56 **(d)** −29

2. For the sequence $t_n = 7 - 2n$, the value of t_{15} is: 1

(a) −23 **(b)** −46 **(c)** −22 **(d)** −24

3. Which term of the sequence $t_n = 4n - 3$ is 97? 1

(a) $n = 26$ **(b)** $n = 24$ **(c)** $n = 25$ **(d)** $n = 27$

4. The 9th term of the AP $11, 8, 5, \ldots$ is: 1

(a) −13 **(b)** −12 **(c)** −26 **(d)** −14

5. The sum of the first 10 terms of the AP with $a = 2,\ d = 3$ is: 1

(a) 156 **(b)** 310 **(c)** 155 **(d)** 154

6. The sum of the first 8 terms of the AP with $a = 5,\ d = 5$ is: 1

(a) 181 **(b)** 179 **(c)** 360 **(d)** 180

7. The 4th term of the GP with first term 4 and common ratio 3 is: 1

(a) 109 **(b)** 108 **(c)** 216 **(d)** 107

8. In the Virahānka–Fibonacci sequence $1, 2, 3, 5, 8, 13, \ldots$ the 7th term is: 1

(a) 34 **(b)** 21 **(c)** 20 **(d)** 18

9. The sequence $1, 4, 9, 16, \ldots$ is: 1

(a) neither an AP nor a GP **(b)** a GP

(c) an AP **(d)** both an AP and a GP

10. In a geometric progression, the constant multiplier is called the: 1

(a) nth term **(b)** first term

(c) common ratio **(d)** common difference

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Determine whether 97 and 172 are terms of $t_n = 5n - 3$. 2

12. Which term of $t_n = 5n - 3$ is 607? 2

13. A sequence is given by $t_1 = -5$, $t_{n+1} = t_n + 3$. Write the first five terms and decide whether 52 is a term. 2

14. Let $T_1 = 1, T_2 = 2, T_3 = 4$ and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$. Find T_4, T_5, T_6. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find $12 + 13 + 14 + \cdots + 40$ using $S_n = \frac{n(n+1)}{2}$. 3

16. Find the sum of the first 20 terms of the AP $3, 8, 13, \ldots$ using 'average of first and last × number of terms'. 3

17. For the sequence $1, 4, 7, 10, \ldots$ write the explicit rule and the recursive rule. 3

OR

Find the 10th, 17th and 80th triangular numbers. 3

18. Verify that $1 + 3 + 5 + 7 + 9 = 5^2$ and prove that the sum of the first n odd numbers is n^2 using the AP sum. 3

OR

Find the 10th term of the GP $3, 6, 12, \ldots$ and its recursive rule. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A ball is dropped from 24 ft and rebounds to $\frac{3}{4}$ of its previous height each time. 5

(i) Find the heights after the first five bounces.
(ii) Write the nth term.
(iii) After how many bounces does the height first fall below 3 ft?

OR

Case study: Harish's starting salary is ₹5,00,000 and he gets an increment of ₹20,000 each year. 5

(i) Write his salary in the first three years.
(ii) Find the salary in the 8th year.
(iii) In which year will it be ₹8,00,000?
(iv) Find his total earnings in 10 years.

20. An AP has 50 terms with $t_3 = 12$ and $t_{50} = 106$. 5

(i) Find a and d.
(ii) Find the 29th term.
(iii) Find the sum of all 50 terms.

OR

(i) Write the first ten terms of the Virahānka–Fibonacci sequence. 5
(ii) Find the sum of the first 8 terms and compare with V_{10}.
(iii) State where and when it was first written down.

END OF QUESTION PAPER — Total: 40 Marks